



SCHOOL of DATA SCIENCE

Design a Unique ID Generator

Distributed Computing
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The Problem

Consider: Database table is partitioned and placed on multiple servers (distributed system)

Need to generate and store unique session IDs or user IDs

```
user_session_id
11111111111111
11111111111112
11111111111113
11111111111114
```

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Auto Increment

A database has an auto increment function that can generate sequential numbers.

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Since the data is distributed, this won't work...
there can be duplicates across the servers

server1

```
user_session_id
11111111111111
11111111111112
11111111111113
11111111111114
```

server2

```
user_session_id
11111111111111
11111111111112
11111111111113
11111111111114
```

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UUIDs

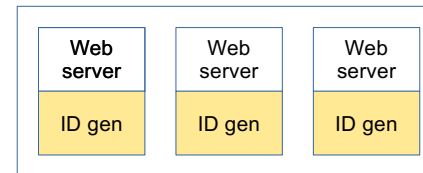
One potential solution – universally unique identifier (UUID)

128-bit number used to identify information in computer systems

Very unlikely to create duplicates

UUIDs can be generated independently without coordination between servers

UUID generator



Each server has an ID generator. They operate independently.

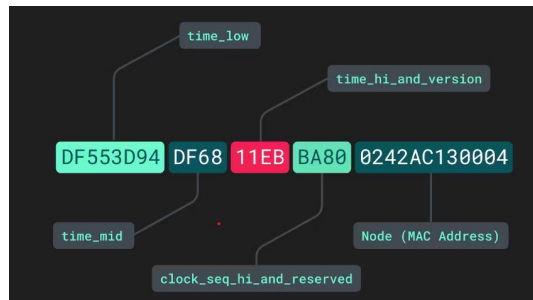
UUID examples

```
6ddc8513-dc7b-3b37-b21b-a1ca9440fe14
8d6a1314-170a-559c-afe7-b68d1d7ee9ac
```

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Anatomy of UUID



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UUID Generation

UUID is a subtype of the *string* data type

```
CREATE TABLE users (  
  id UUID NOT NULL DEFAULT gen_random_uuid(),  
  city STRING NOT NULL,  
  name STRING NULL,  
  address STRING NULL,  
  credit_card STRING NULL,  
  CONSTRAINT "primary" PRIMARY KEY (city ASC, id ASC),  
  FAMILY "primary"  
(id, city, name, address, credit_card)  
);
```

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References

System design interview. Alex Xu.

<https://www.mparticle.com/blog/what-is-a-uuid/>